

Telecom Customer Churn Analysis Report

Project Overview

This analysis was conducted to identify the key drivers of customer churn, estimate the business revenue impact, and recommend strategic actions to improve customer retention for a telecom business.

The dataset included customer demographics, subscription behavior, contract details, service usage, billing information, and churn status.

Executive Summary

The analysis identified several strong churn indicators directly impacting customer retention and recurring revenue.

Key Findings

- Total Customers Analyzed: **7,043**
 - Total Churned Customers: **1,869**
 - Overall Churn Rate: **26.54%**
 - Estimated Monthly Revenue Lost Due to Churn: **₹139,130.85**
 - Nearly **1 in 4 customers** are leaving the business.
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Detailed Business Insights

1. Contract Type is the Strongest Driver of Churn

Observation

Customers with **Month-to-Month contracts** showed the highest churn behavior compared to customers on long-term contracts.

Revenue Impact

Contract Type	Revenue Lost	Contribution
Month-to-Month	₹120,847.10	86.86%
One Year	₹14,118.45	10.15%
Two Year	₹4,165.30	2.99%

Business Insight

- Short-term customers are significantly less loyal.
- Long-term contracts create stronger customer retention and predictable recurring revenue.

Recommendation

- Introduce discounts for annual subscriptions.
 - Offer loyalty rewards for contract upgrades.
 - Create retention campaigns targeting month-to-month customers.
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2. High Monthly Charges Increase Churn Probability

Observation

Customers who churned had significantly higher monthly bills.

Customer Status	Average Monthly Charges
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Active Customers ₹61.27

Churned
Customers ₹74.44

Business Insight

- Customers paying higher monthly charges are more price-sensitive.
- Premium pricing without perceived value may be increasing churn risk.

Recommendation

- Create bundled service packages.
 - Offer personalized discounts to high-paying customers.
 - Improve value perception through premium support and loyalty benefits.
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3. New Customers are More Likely to Churn

Observation

Customers with lower tenure (1–2 months) showed the highest churn rates.

Business Insight

- Customer onboarding and early engagement are weak.
- The business is failing to build long-term relationships during the initial customer lifecycle.

Recommendation

- Improve onboarding experience.
 - Introduce first-90-day engagement campaigns.
 - Provide welcome offers and proactive customer support.
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4. Senior Citizens Show Higher Churn Rates

Observation

Senior citizen customers showed comparatively higher churn percentages than non-senior customers.

Business Insight

- Older customers may face usability, pricing, or support-related challenges.

Recommendation

- Simplify service plans.
 - Improve customer support accessibility.
 - Provide senior-citizen-focused service packages.
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5. Fiber Optic Customers Generate the Highest Revenue Leakage

Revenue Loss by Internet Service

Internet Service	Revenue Lost
Fiber Optic	₹114,300.05
DSL	₹22,529.20
No Internet	₹2,301.60

Key Insight

- Fiber optic customers contributed **82.15%** of total churn-related revenue loss.
- High-paying premium customers are leaving at a higher rate.

Recommendation

- Improve fiber network quality and support experience.
 - Launch premium customer retention programs.
 - Monitor complaints and service downtime closely.
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6. Electronic Check Users Have Higher Churn Tendencies

Observation

Customers using **Electronic Check** payment methods showed noticeably higher churn behavior.

Business Insight

This may indicate:

- Payment friction
- Billing dissatisfaction
- Lower customer trust

Recommendation

- Encourage auto-pay and credit card billing.
 - Offer cashback or incentives for stable payment methods.
 - Improve billing transparency.
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7. Lack of Support Services Correlates with Churn

Observation

Customers without:

- Online Security
- Tech Support
- Online Backup
- Device Protection

showed significantly higher churn behavior.

Business Insight

Customers using additional support services are more engaged and retained longer.

Recommendation

- Bundle support services with internet plans.
 - Promote value-added services during onboarding.
 - Cross-sell protection and support packages.
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High-Value Customer Churn Analysis

Observation

- High-value churn customers (Monthly Charges > ₹80): **906 customers**
- Contribution among churned customers: **48.48%**

Business Insight

Nearly half of churned customers are premium-paying customers, creating major revenue leakage.

Recommendation

- Build VIP retention strategies.
 - Introduce premium customer loyalty programs.
 - Create proactive churn prediction alerts for high-value users.
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Strategic Recommendations

Immediate Actions (0–3 Months)

- Target month-to-month customers with retention offers.
 - Reduce churn among fiber optic users.
 - Improve onboarding for new customers.
 - Launch customer support campaigns.
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Medium-Term Actions (3–6 Months)

- Introduce loyalty and rewards programs.
- Bundle value-added services.
- Improve pricing optimization strategies.